

BLOCKCHAIN & CYBERSECURITY

SOFTWARE PROJECT

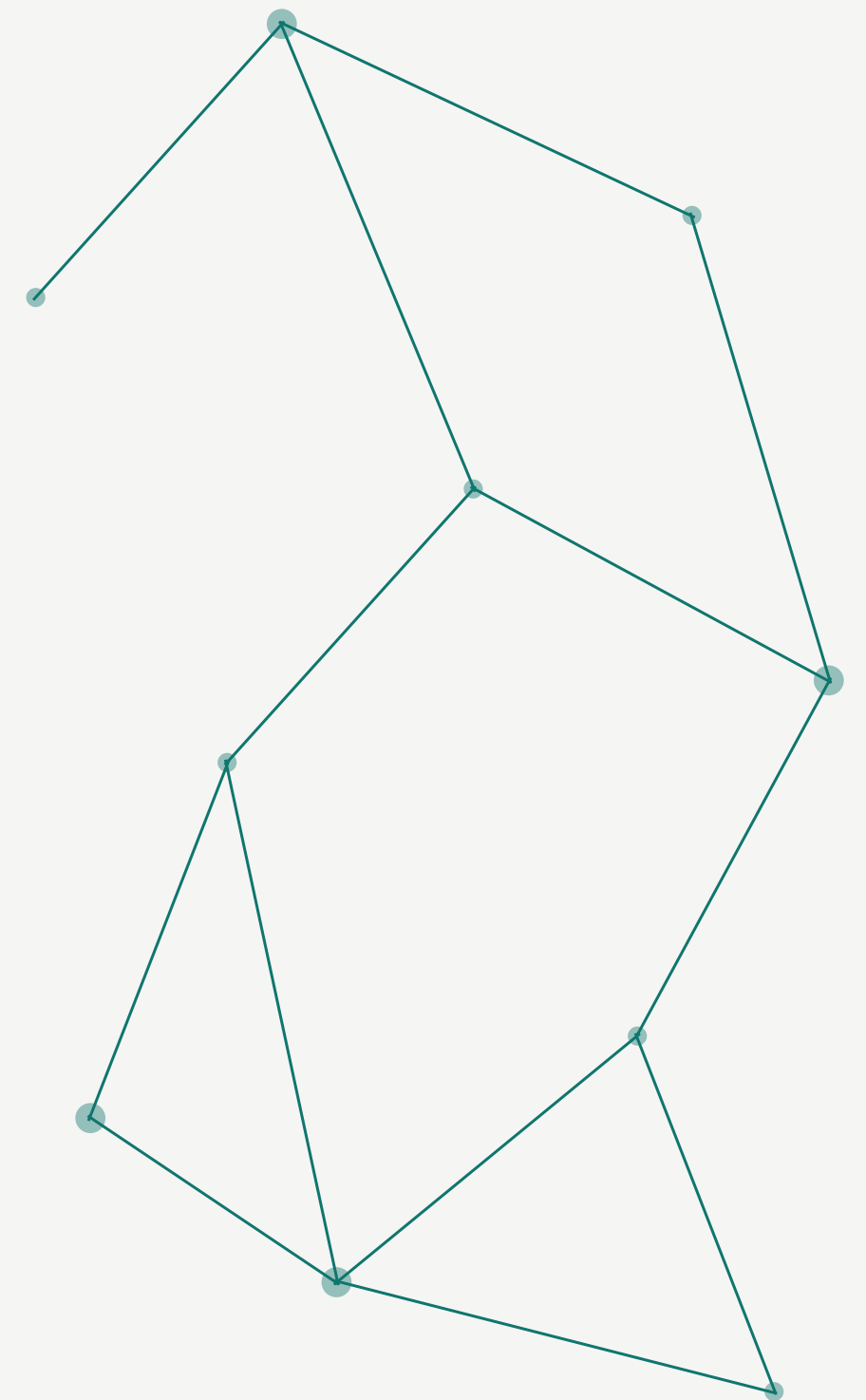
Blockchain-Based Academic Degree Verification & Credential Authentication

GUIDE

Ms. Chetana Bhagat

PRESENTED BY

Ramsha Siddiqui • Aditya Dhakarge
Tukaram Magar • Priyatam More



Why This Project Exists

PROBLEM

Problem Statement

- Forged degrees and manipulated transcripts undermine trust in academic qualifications.
- Manual, document-based verification is slow and error-prone.

NEED

Need of the Project

- No secure, transparent, universally accessible way exists to verify credentials quickly.
- Institutions and employers rely on slow registrar communication.

GOALS

Objectives

- Eliminate fake-degree fraud and build trust between universities and employers.
- Create permanent, tamper-proof academic records with instant verification.

APPROACH

Purpose

- Universities issue certificates digitally; proof is stored on blockchain.
- Employers verify instantly via QR code or certificate ID.

BEFORE

Traditional Verification

Manual, registrar-dependent, slow



AFTER

Blockchain Verification

Instant, tamper-proof, self-service

Comparative Study of Existing Systems

Platform	Blockchain Used	Strength	Limitation
Blockcerts	Bitcoin / Ethereum	Open-source; widest institutional adoption worldwide	Public chain – hash-only, limited privacy controls
OpenCerts	Ethereum	Government-mandated; proven at national scale (2M+ certs)	Dependent on Ethereum network fees and congestion
EduCTX	Ark Blockchain	First ECTS-compatible blockchain credit platform	Piloted only; not yet deployed at production scale
Sony Global Ed.	Hyperledger Fabric	GDPR-aligned, privacy-by-design architecture	Permissioned network – limited public verifiability
EBSI	Permissioned (EU nodes)	Cross-border legal framework; W3C VC compliant	Multi-jurisdiction governance remains complex

RESEARCH GAPS IDENTIFIED

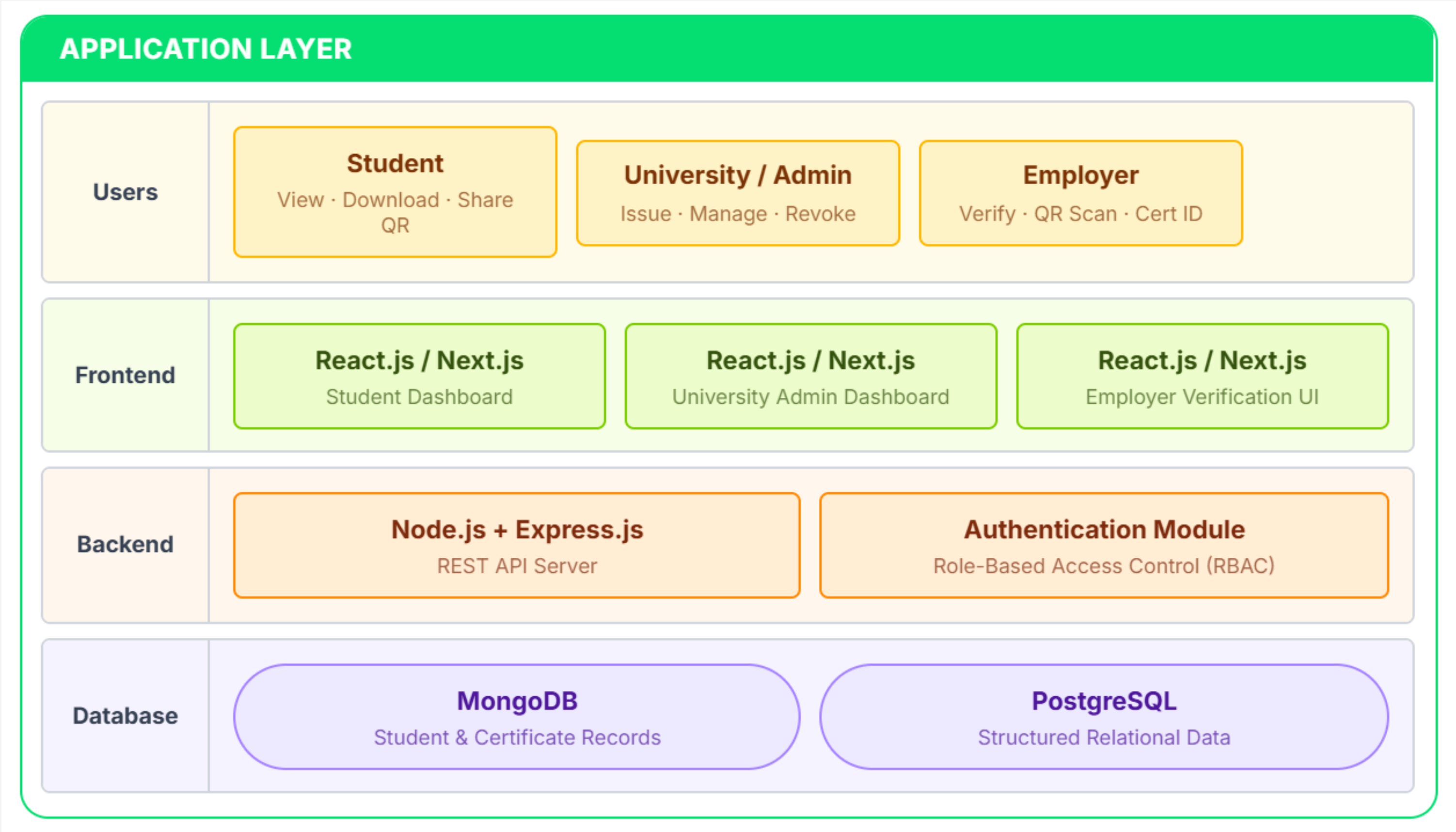
No Universal Standard

Privacy vs Immutability

Scalability

Digital Inclusion

End-to-End Architecture



BLOCKCHAIN & VERIFICATION LAYER

Blockchain Layer

Ethereum / Polygon

Decentralized Blockchain Network

Smart Contracts

Issuance · Validation · Revocation

Certificate Hash Storage

Immutable On-Chain Record

Verification Layer

QR Verification

Certificate ID / QR Scan

Employer Verification

Instant Self-Service Check

Revocation Check

Real-Time Status on Chain

ISSUANCE & VERIFICATION WORKFLOW

1. University Login



2. Enter Student Details



3. Generate Certificate



4. SHA-256 Hash



5. Store on Blockchain



6. Issue to Student



7. Student Shares QR



8. Employer Scans



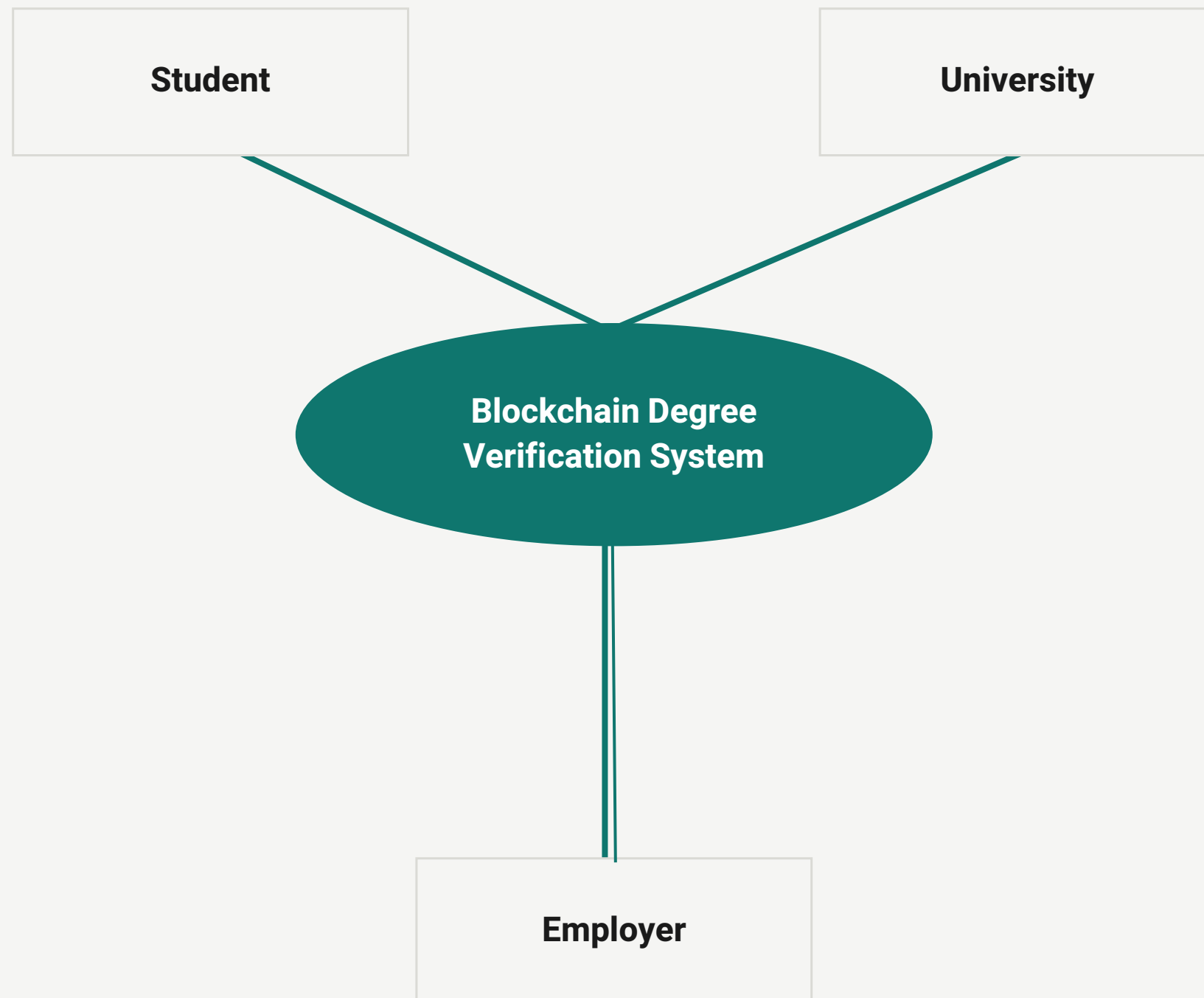
9. Blockchain Validates



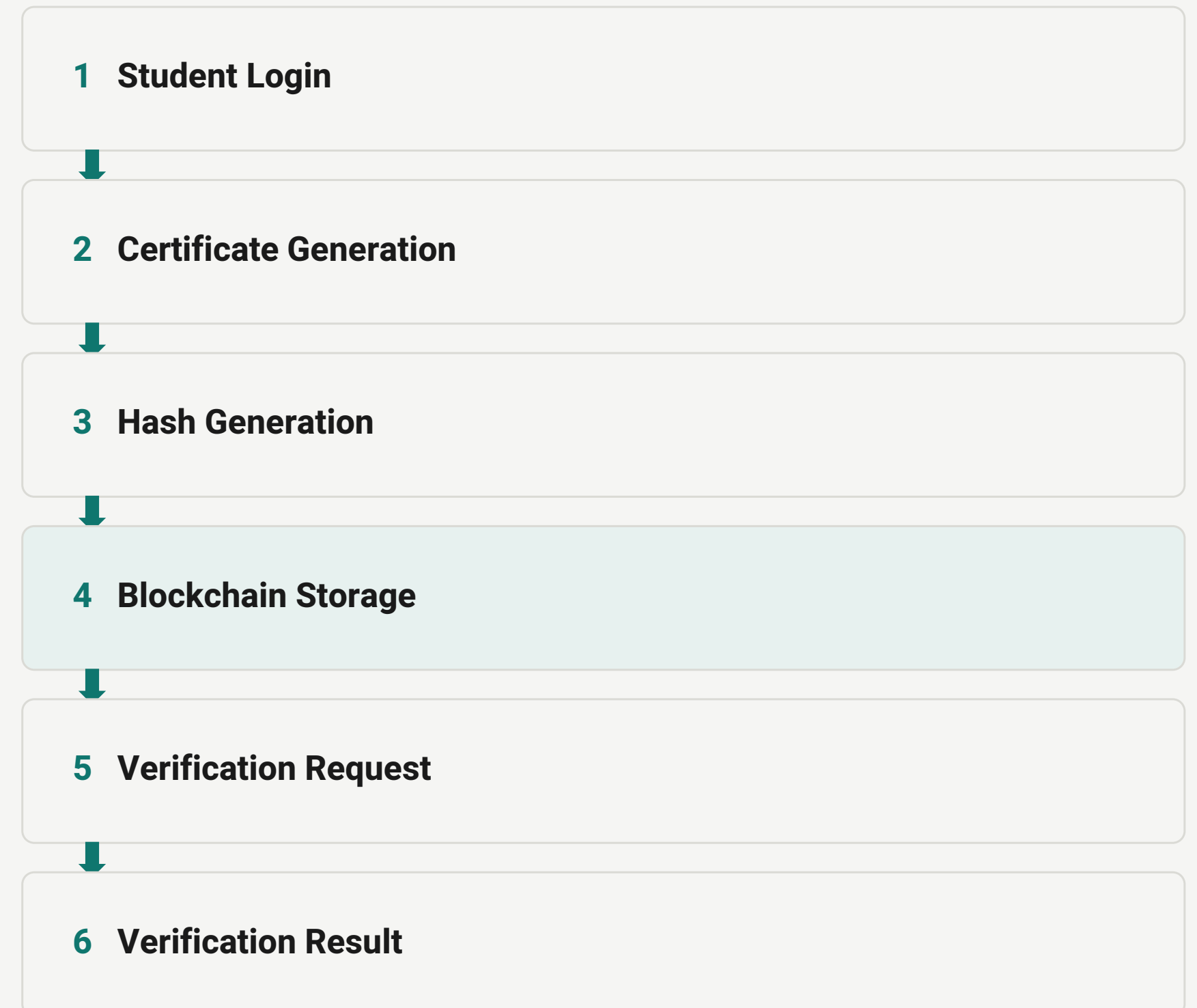
10. Verified ✓

DFD – Level 0 & Level 1

LEVEL 0 – CONTEXT DIAGRAM

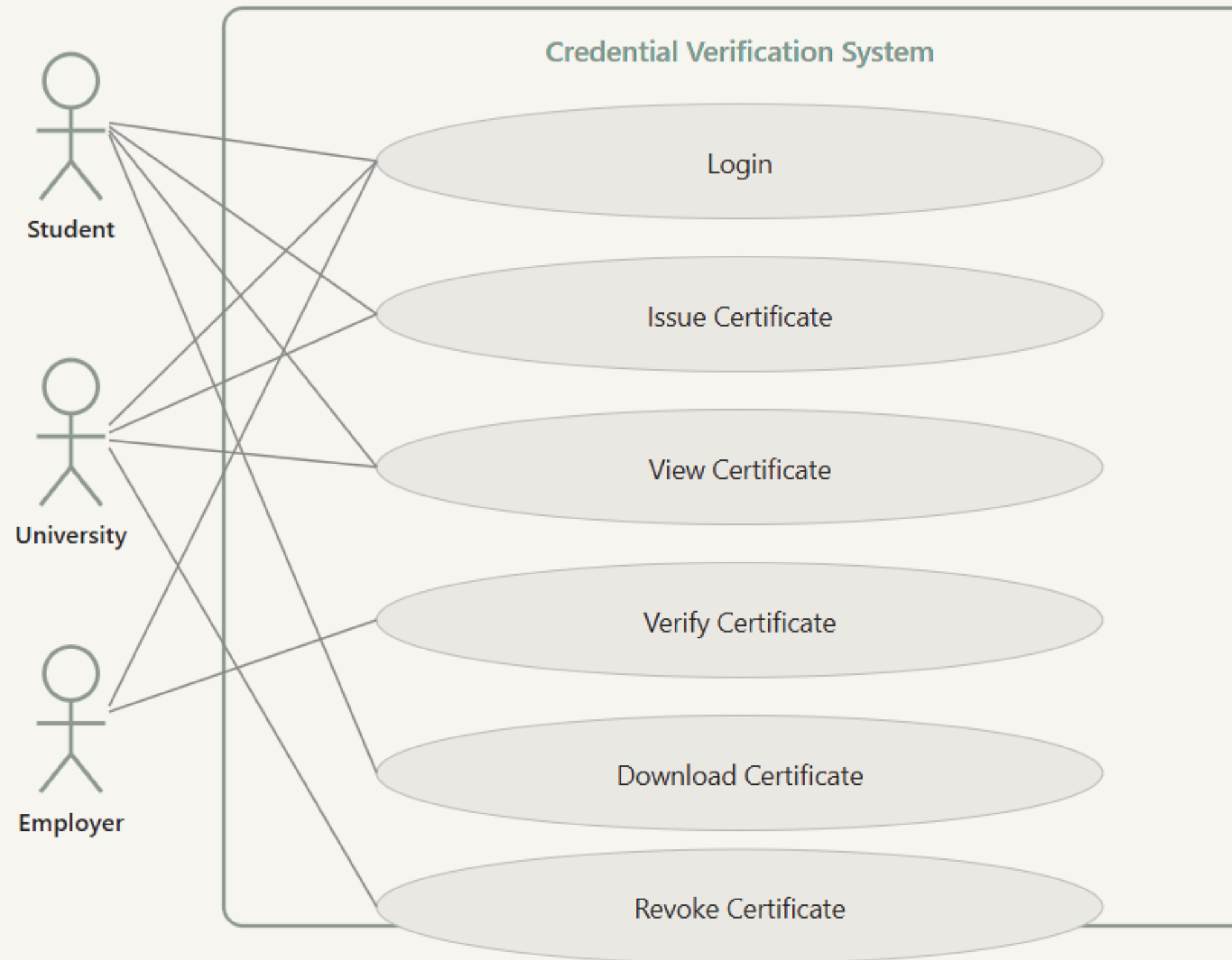


LEVEL 1 – PROCESS FLOW

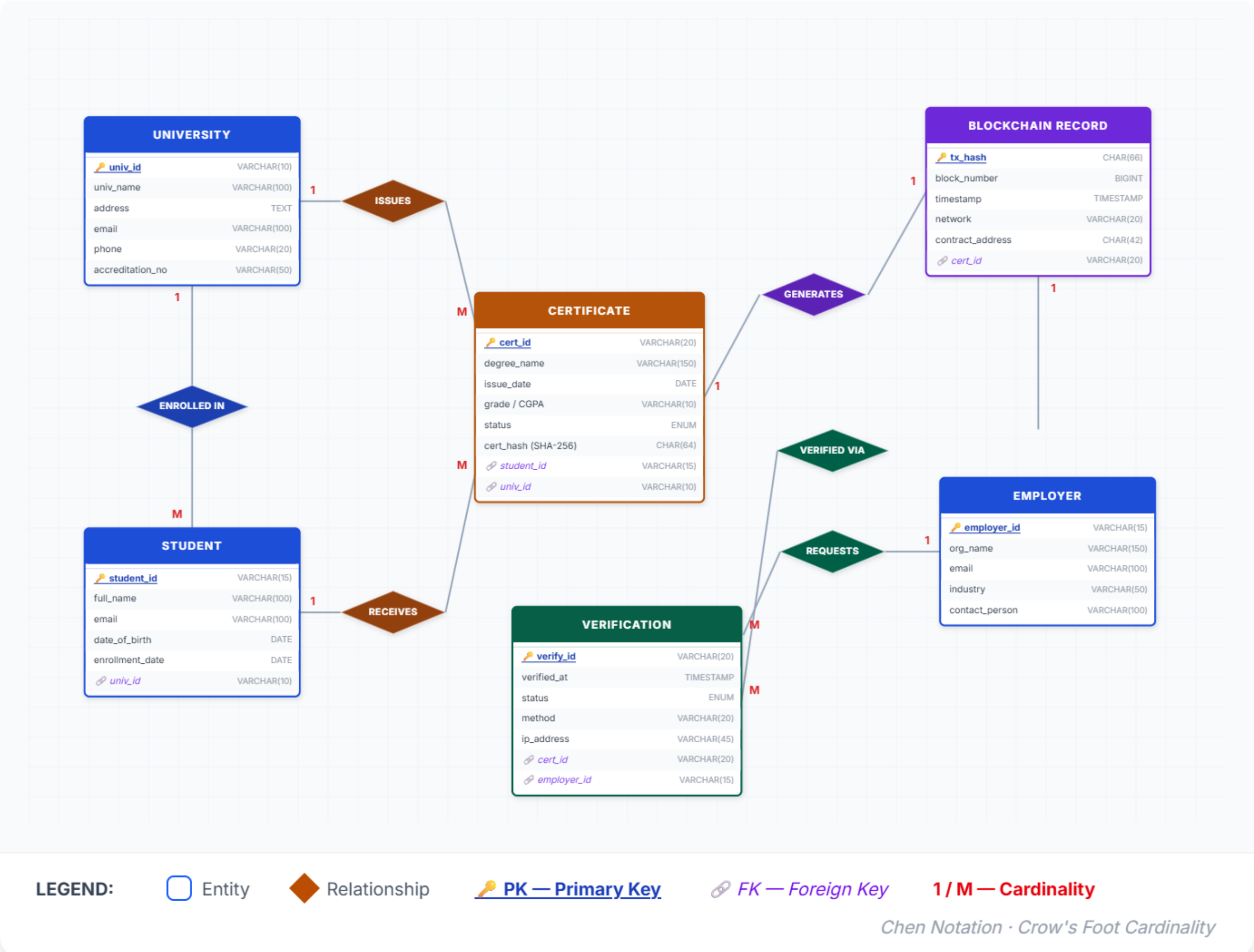


Use Case & Entity Relationship Diagrams

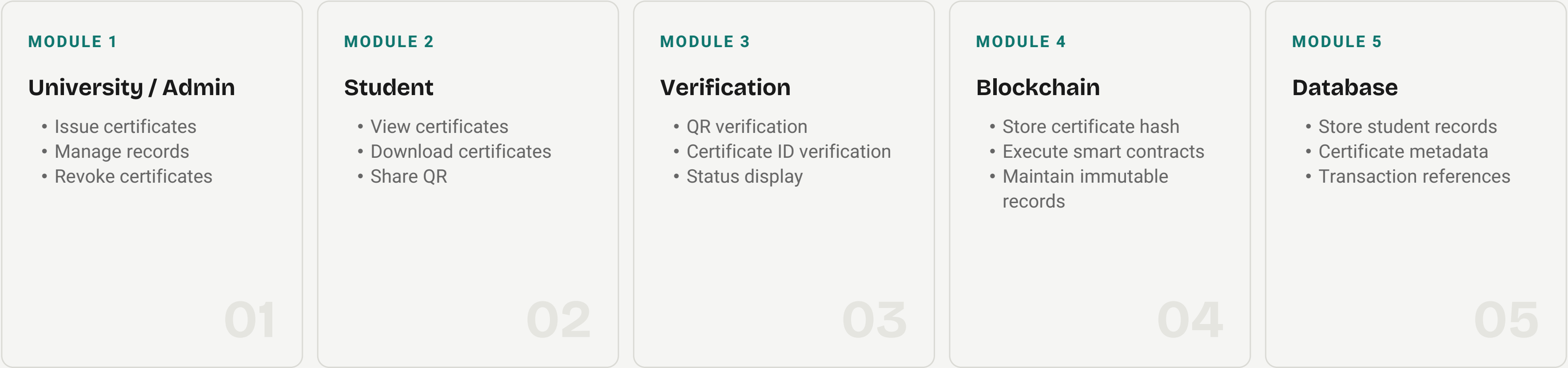
Use Case Diagrams



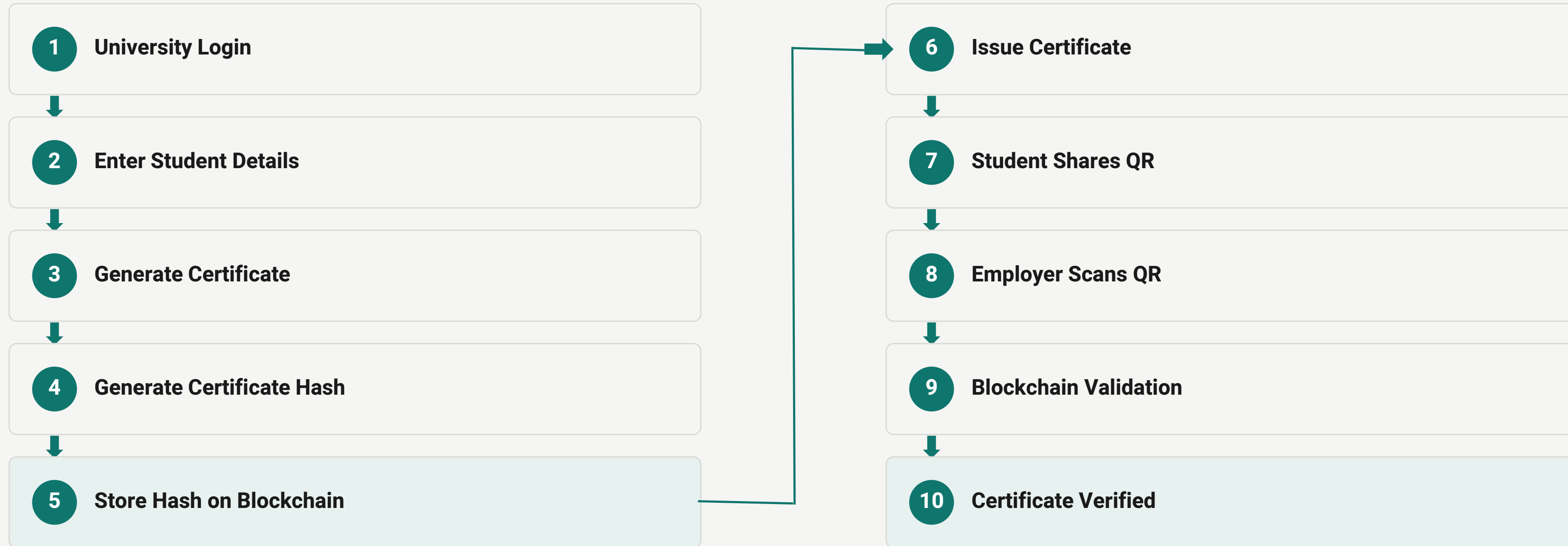
ER Diagram



System Modules



Issuance & Verification Workflow



Achievements & Future Scope

ACHIEVEMENTS

1 Tamper-proof verification

2 Instant authentication

3 Reduced fraud

4 QR verification

5 Secure blockchain storage

FUTURE SCOPE

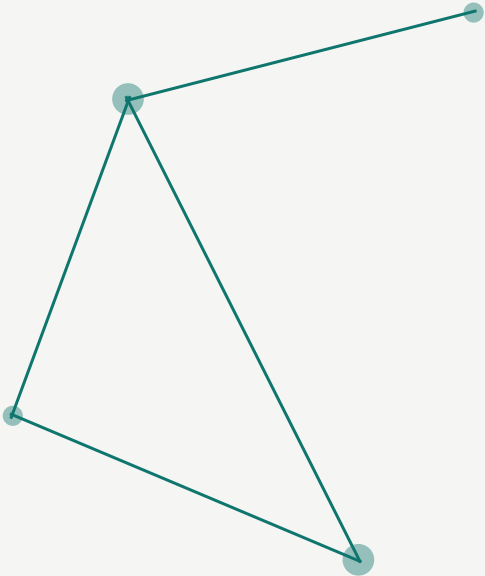
1 DigiLocker integration

2 Government board support

3 International universities

4 NFT credentials

5 Mobile application



Thank You

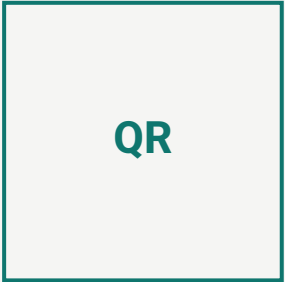
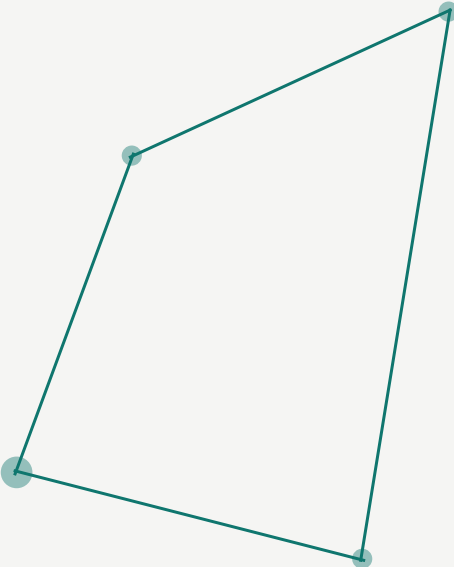
Questions?

GUIDE

Ms. Chetana Bhagat

TEAM MEMBERS

Ramsha Siddiqui • Aditya Dhakarge
Tukaram Magar • Priyatam More



QR